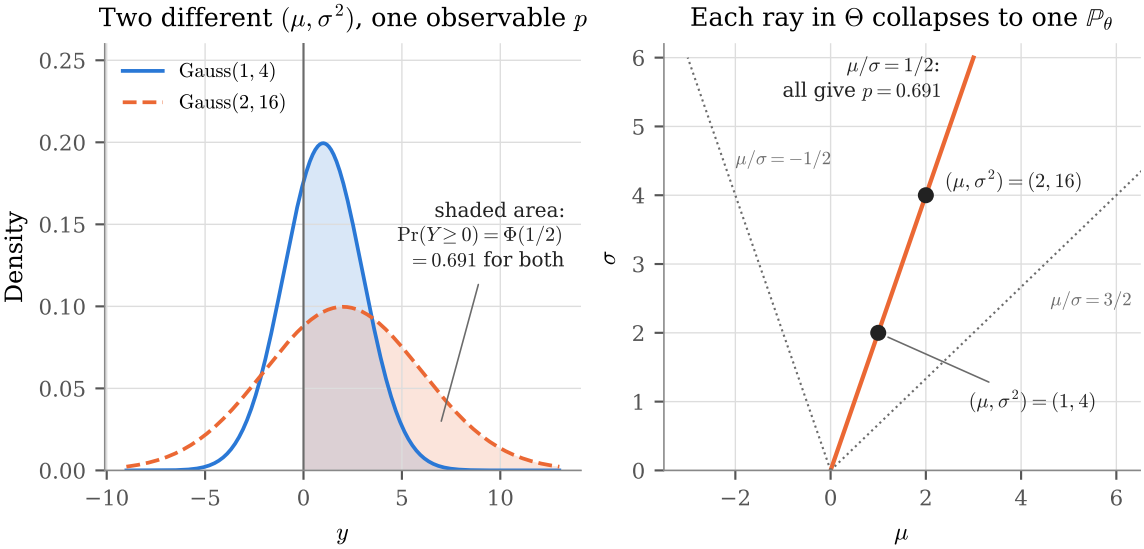
Two different (μ, σ^2), one observable p
Gauss(1, 4)
Gauss(2, 16)
shaded area:
$\Pr(Y \geq 0) = \Phi(1/2)$
$= 0.691$ for both
Density
0.25
0.20
0.15
0.10
0.05
0.00
−10
−5
0
5
10
y
Each ray in Θ collapses to one $\mathbb{P}_\theta$
$\mu/\sigma = 1/2$:
all give $p = 0.691$
$\mu/\sigma = -1/2$
$(\mu, \sigma^2) = (2, 16)$
$\mu/\sigma = 3/2$
$(\mu, \sigma^2) = (1, 4)$
σ
6
5
4
3
2
1
0
−2
0
2
4
6
μ